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9618/21

May/June 2024

2 hours

You must answer on the question paper.

You will need: Insert (enclosed)

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].
- No marks will be awarded for using brand names of software packages or hardware.
- The insert contains all the resources referred to in the questions.

This document has **24** pages. Any blank pages are indicated.

Refer to the **insert** for the list of pseudocode functions and operators.

1 An algorithm is developed in pseudocode before being coded in a programming language.

(a) The following table shows four valid pseudocode assignment statements.

Complete the table by giving an appropriate data type to declare each of the variables A, B, C and D.

Assignment statement	Data type
A ← LEFT(MyName, 1)	
B ← Total * 2	
C ← INT(ItemCost) / 3	
D ← "Odd OR Even"	

[4]

(b) Other variables in the program have example values as shown:

Variable	Value
Sorted	False
Tries	9
ID	"ZGAC001"

Complete the table by evaluating each expression, using the example values.

Expression	Evaluates to
Tries < 10 AND NOT Sorted	
Tries MOD 4	
TO_LOWER(MID(ID, 3, 1))	
LENGTH(ID & "xx") >= Tries	

[4]

(c) The variable names A, B, C and D in part (a) are **not** good programming practice.

(i) State why these variable names are **not** suitable.

.....
..... [1]

(ii) Identify **one** problem that these variable names might cause.

.....
..... [1]

(iii) The choice of suitable variable names is one example of good programming practice.

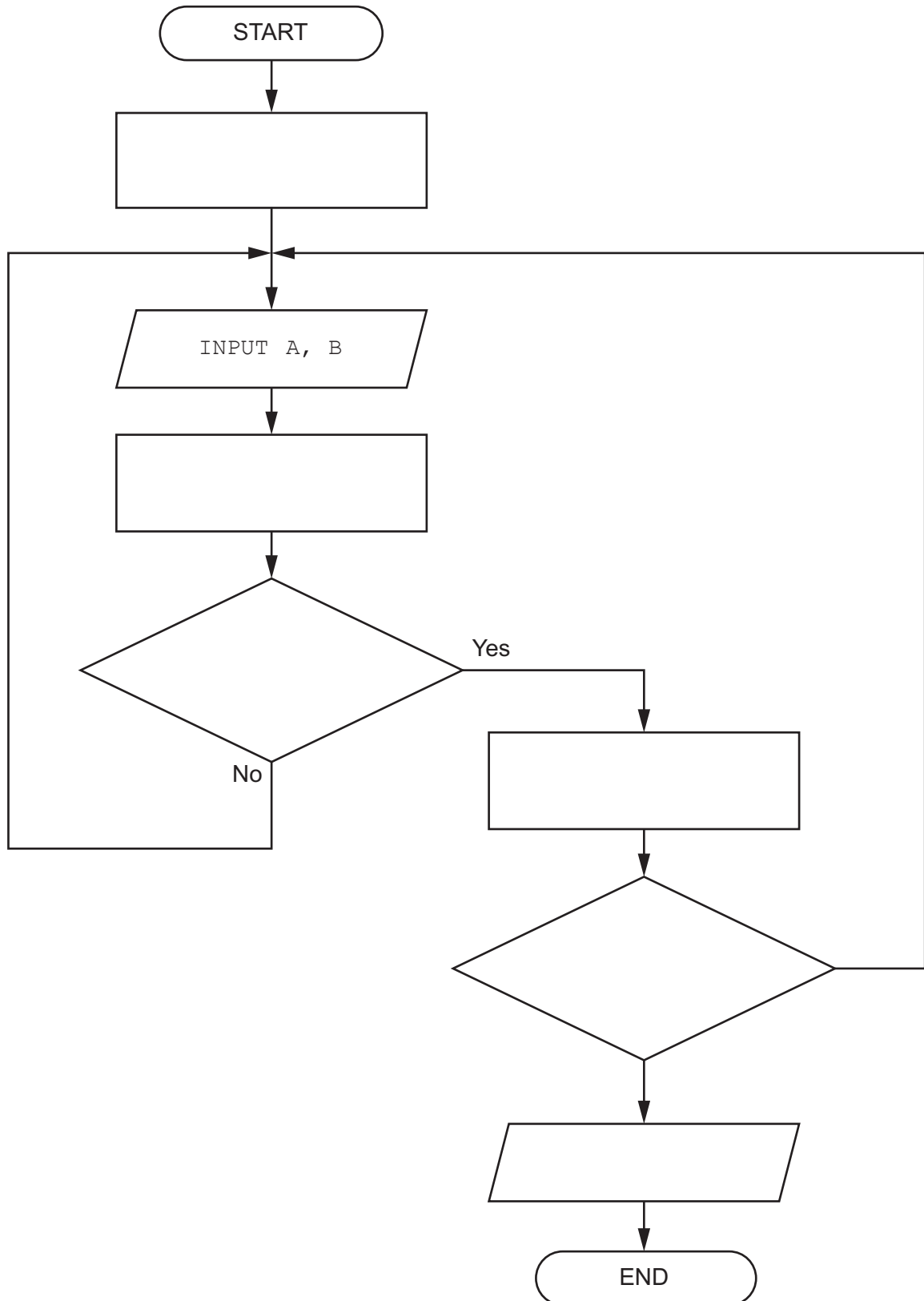
Give **one other** example.

.....
..... [1]

2 An algorithm has three steps. It will:

1. repeatedly input a pair of numeric values A and B
2. count the number of pairs that are input until A has been greater than B 10 times
3. output the number of pairs that were input.

(a) Complete the program flowchart.



(b) Step 1 of the algorithm is changed.

A variable `ThisSequence` is used to enter a sequence of 10 pairs of numeric values, using a single input statement.

Following the input of `ThisSequence` the revised algorithm will extract the pairs of numbers.

Describe the variable `ThisSequence` and how the numbers are extracted.

.....

.....

.....

..... [2]

3 The diagram shows an Abstract Data Type (ADT) representation of a linked list after data items have been added.

- PS is the start pointer.
- PF is the free list pointer.
- Labels Df, Dc, Db and Dy represent the data items of nodes in the list.
- Labels Fg, Fh, Fm and Fw represent the data items of nodes in the free list.
- The symbol $\emptyset$ represents a null pointer.

PS



PF



(a) Describe the linked list immediately after initialisation, before **any** data items are added.

.....

.....

.....

.....

.....

..... [3]

- (b) A program will be written to include a linked list to store alphanumeric user IDs.

The design uses two variables and two 1D arrays to implement the linked list.
Each array element contains data of a single data type and **not** a record.

The statements below describe the design.

Complete the statements.

The two variables will be of type

The two variables will be used as to the arrays.

The values stored in the two variables will indicate

.....

The first 1D array will be of type

The first 1D array will be used to

The second 1D array will be of type

The second 1D array will be used to

[5]

A function `Check()` will:

- Write pseudocode for the function `Check()`.

..... [6]

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- 5 A global 1D array of strings contains three elements which are assigned values as shown:

```
Data[1] ← "aaaaaa"
Data[2] ← "bbbbbb"
Data[3] ← "cccccc"
```

Procedure `Process()` manipulates the values in the array.

The procedure is written in pseudocode as follows:

```
PROCEDURE Process (Format : STRING)
    DECLARE Count, Index, L : INTEGER
    DECLARE Result : STRING
    DECLARE C : CHAR

    Result ← "*****"

    FOR Count ← 1 TO LENGTH(Format) STEP 2
        C ← MID(Format, Count, 1)
        L ← STR_TO_NUM(MID(Format, Count + 1, 1))

        Index ← (Count + 1) DIV 2

        CASE OF C
            'X' : Result ← TO_UPPER(Data[Index])
            'Y' : Result ← TO_LOWER(Data[Index])
            'Z' : Result ← "***" & Data[Index]
        ENDCASE

        Data[Index] ← LEFT(Result, L)
    NEXT Count

ENDPROCEDURE
```

- (a) Complete the trace table by dry running the procedure when it is called as follows:

CALL Process ("X3Y2W4")

Count	C	L	Index	Result	Data [1]	Data [2]	Data [3]

[6]

- (b) The procedure is to be modified. If variable C is assigned a value other than 'X', 'Y' or 'Z', then procedure Error() is called and passed the value of variable C as a parameter.

This modification can be implemented by adding a **single line** of pseudocode.

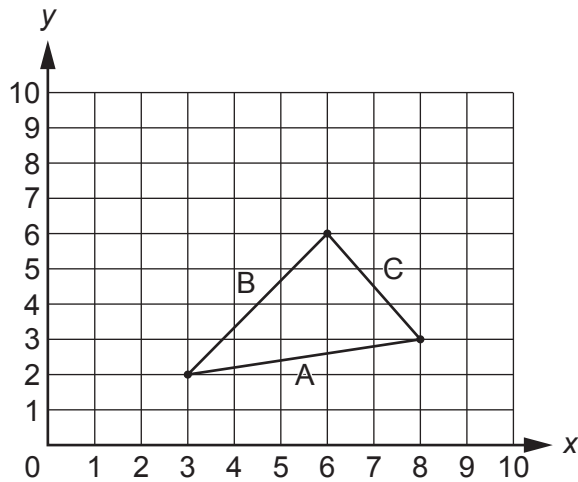
- (i) Write the single line of pseudocode.

..... [1]

- (ii) State where this new line should be placed.

..... [1]

- 6 Three points on a grid form a triangle with sides of length A, B and C as shown in the example:



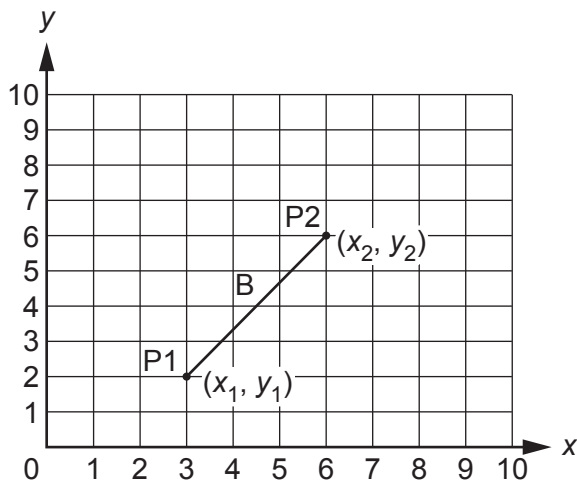
A triangle is said to be right-angled if the following test is true (where A is the length of the longest side):

$$A^2 = B^2 + C^2$$

A^2 means A multiplied by A, for example 3^2 means 3×3 which evaluates to 9

You can calculate A^2 , B^2 and C^2 by using the coordinates of the endpoints of each line.

For example, B^2 is calculated as follows:



The endpoints, P1 and P2, have the coordinates (3, 2) and (6, 6).

The value B^2 is given by the formula:

$$B^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$$

In this example:

$$B^2 = (3 - 6)^2 + (2 - 6)^2$$

$$B^2 = (-3)^2 + (-4)^2$$

$$B^2 = 9 + 16$$

$$B^2 = 25$$

- take three sets of integers as parameters representing the coordinates of the three endpoints that form a triangle
- return `TRUE` if the endpoints form a right-angled triangle, otherwise return `FALSE`.

Complete the pseudocode for the function `IsRA()`.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[6]

(b) The test used to check if a triangle is right-angled can be written in two ways:

$$A^2 = B^2 + C^2$$

or

$$A = \sqrt{B^2 + C^2}$$

The symbol $\sqrt{}$ represents the square root operation. For example, $\sqrt{81} = 9$

A new function `SQRT()` is written to perform the square root operation. The function takes an integer number as a parameter and returns a positive real value representing the square root of the number.

During testing it is found that the `SQRT()` function returns a value that is only accurate to 4 decimal places.

For example, `SQRT(25)` returns 5.0000125 rather than the correct value of 5.0

The function `ISRA()` from part (a) is modified to use the new `SQRT()` function to test if a triangle is right-angled.

Describe a problem that might occur when using the modified `ISRA()` function **and** suggest a solution that still allows the `SQRT()` function to be used.

Problem

.....

Solution

.....

[2]

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- 7 A fitness club has a computerised membership system. The fitness club offers a number of different exercise classes.

The following information is stored for each club member: name, home address, email address, mobile phone number, date of birth and the exercise(s) they are interested in.

- (a) When an exercise class is planned, a new module will send personalised text messages to each member who has expressed an interest in that exercise. Members wishing to join the class send a text message back. Members may decide **not** to receive future text messages by replying with the message 'STOP'.

The process of abstraction is used to filter out unnecessary information.

- (i) State **one** advantage of applying abstraction to this problem.

.....
 [1]

- (ii) Identify **three** items of information that will be required by the new module. Justify your choices with reference to the given scenario.

Item 1 required

Justification

.....

Item 2 required

Justification

.....

Item 3 required

Justification

.....

[3]

- (iii) Identify **two** operations that would be required to process data when the new module receives a text message back from a member.

Operation 1

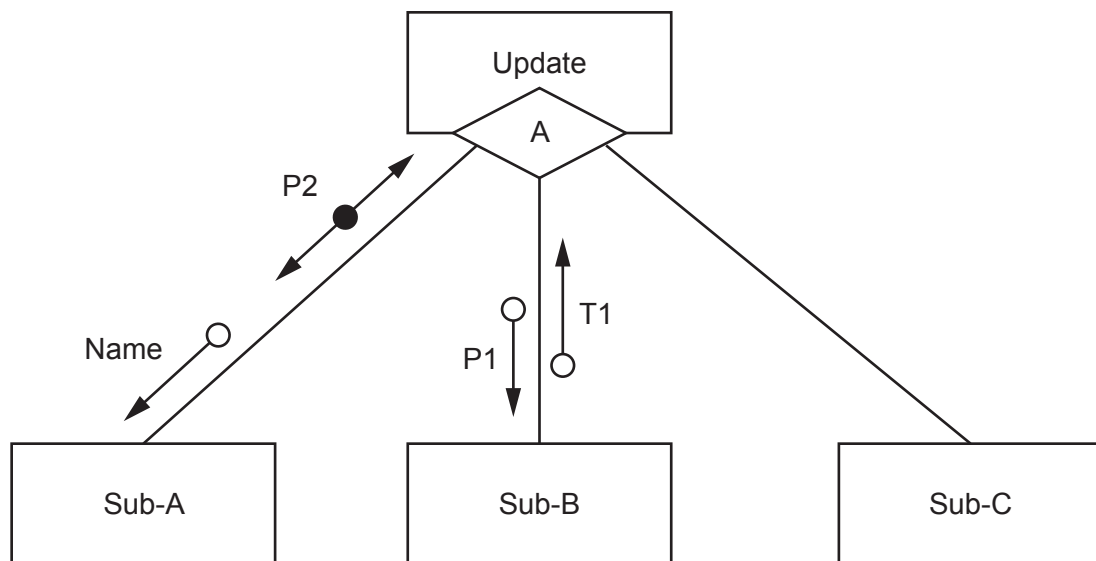
.....

Operation 2

.....

[2]

(b) The structure chart illustrates part of the membership program:



Data item notes:

- Name contains the name of a club member
- P1 and T1 are of type real.

(i) Explain the meaning of the diamond symbol (labelled with the letter A) in the chart.

.....

.....

.....

..... [2]

(ii) Write the pseudocode module headers for Sub-A and Sub-B.

Sub-A

.....

.....

Sub-B

.....

.....

[4]

- 8 A teacher is designing a program to process pseudocode projects written by her students.

Each student project is stored in a text file.

The process is split into a number of stages. Each stage performs a different task and creates a new file named as shown:

File name	Comment
MichaelAday_src.txt	student project file produced by student Michael Aday
MichaelAday_S1.txt	file produced by stage 1
MichaelAday_S2.txt	file produced by stage 2

The teacher has defined the first program module as follows:

Module	Description
DeleteComment()	- called with a parameter of type string representing a line of pseudocode from a student's project file - returns the line after removing any comments Note on comments: A comment starts with two forward slash characters and includes all the remaining characters on the line. The following example shows a string before and after the comment has been removed: Before: IF X2 > 13 THEN //check if limit exceeded After: IF X2 > 13 THEN

(a) Complete the pseudocode for module `DeleteComment()`.

```
FUNCTION DeleteComment(Line : STRING) RETURNS STRING
```

[illegible]

ENDFUNCTION

[8]

(b) A second module is defined:

Module	Description
Stage_1()	- called with a parameter of type string representing a student name - creates a new stage 1 file - copies each line from the student's project file to the stage 1 file after removing any comment from each line - does not write blank lines to the stage 1 file - returns the number of lines written to the stage 1 file

Write pseudocode for module `Stage_1()`.

Module DeleteComment() **must be used in your solution.**

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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..... [7]

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